LAB PROGRAMS 11-21

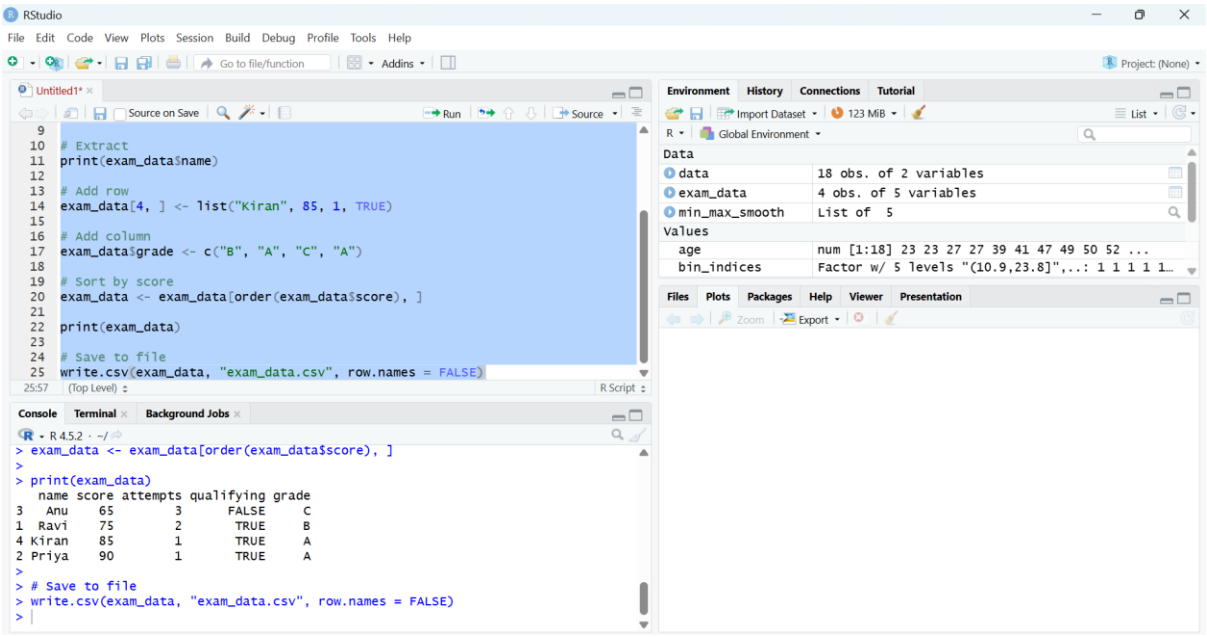
Name: M.Manasa

Reg no: 192425189

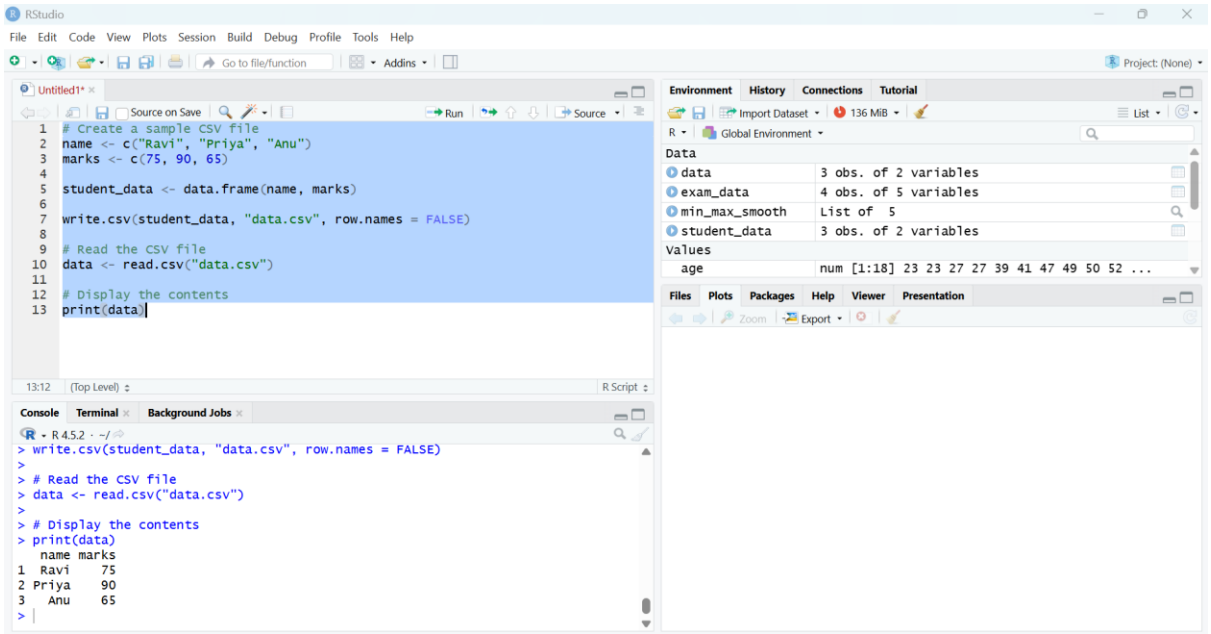
Course code: ITA0408

Course: Statistics with R programming

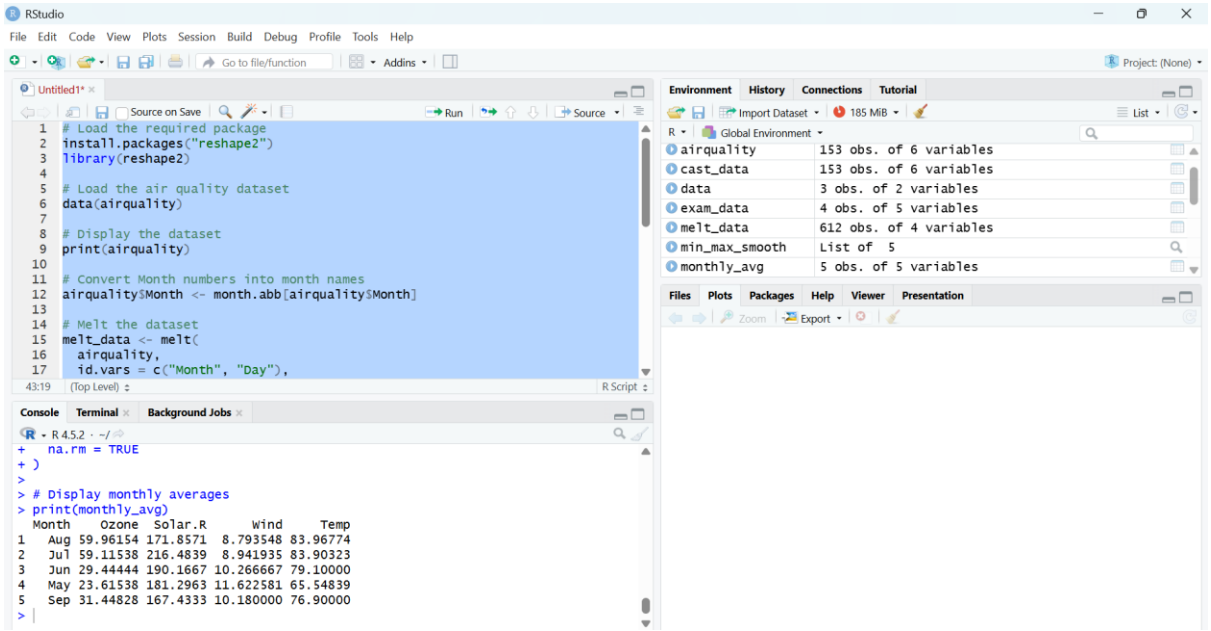
10. Create and explore a data frame exam_data with name, score, attempts, and qualifying fields. Perform extract, add row/column, sort, save to file.



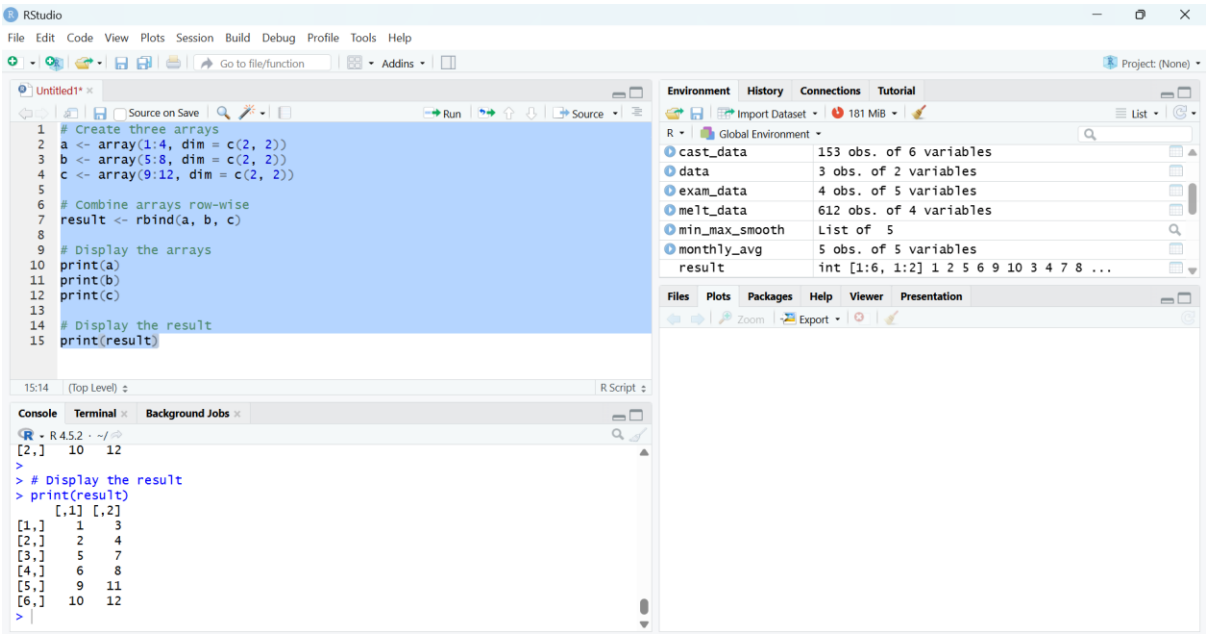
11. Write an R program to read a .csv file and display contents.



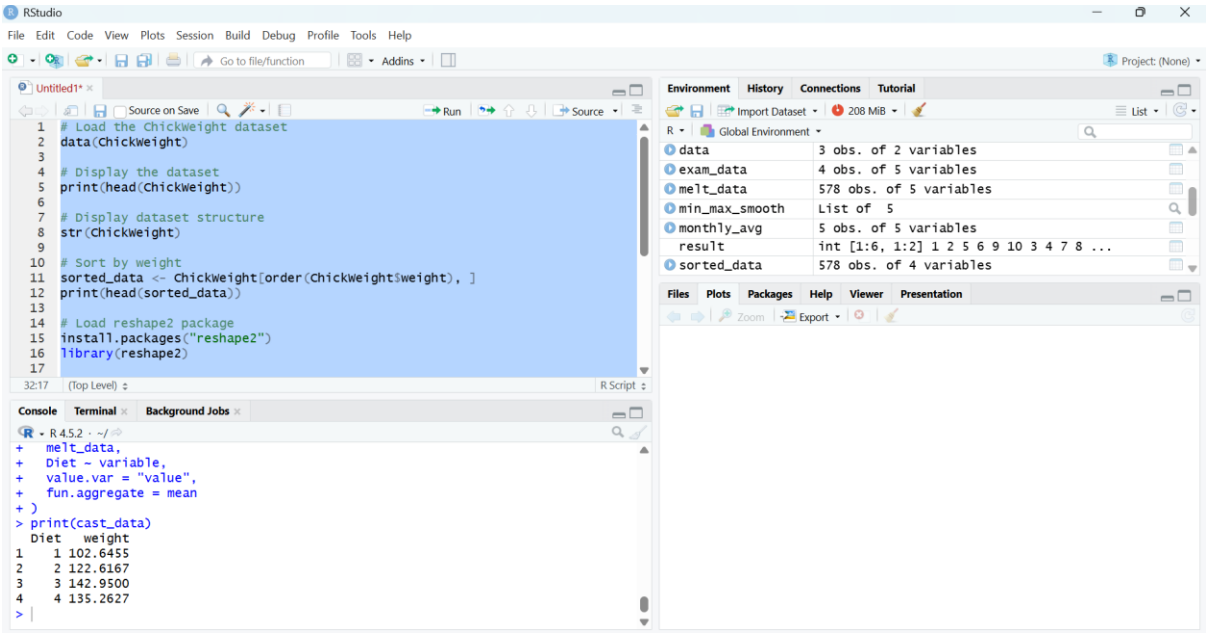
12. Perform data reshaping on air quality dataset: melt, cast, compute monthly averages for Ozone, Solar.R, Wind, and Temperature.



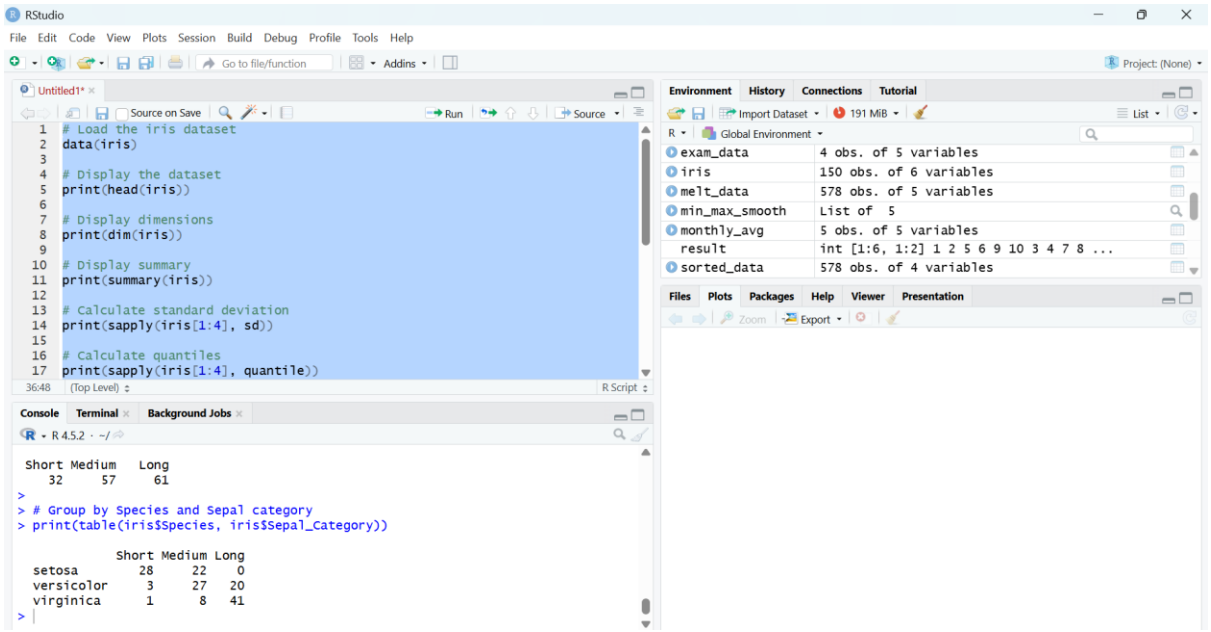
13. Combine multiple arrays row-wise.



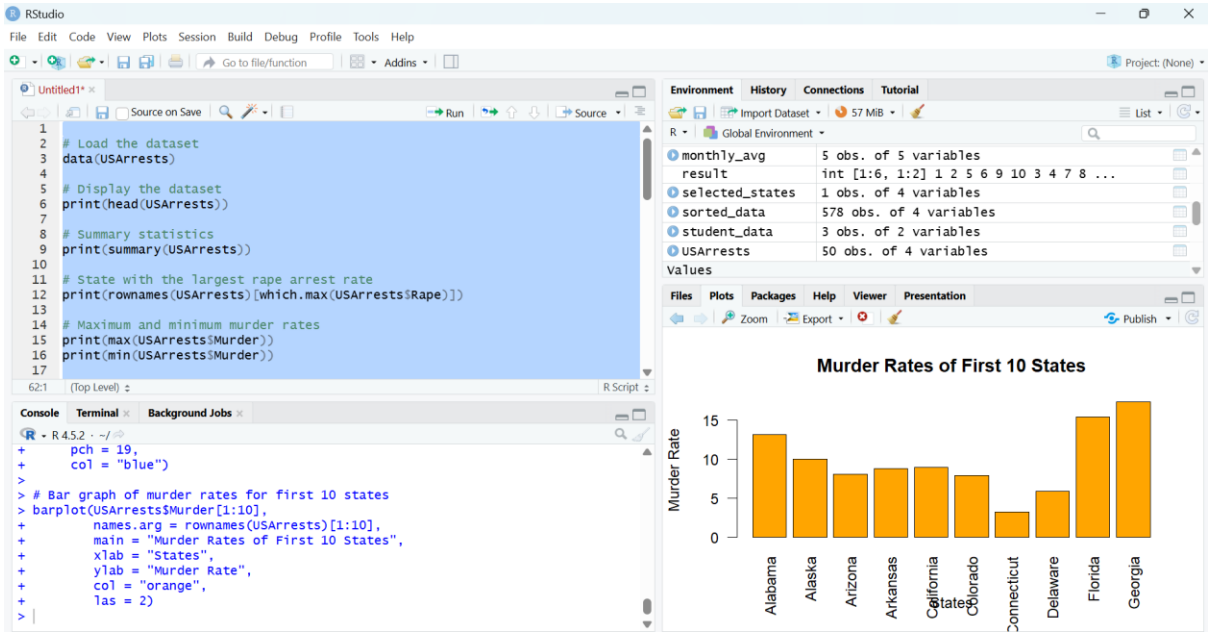
14. Explore and manipulate ChickWeight dataset (sorting, melting, casting by Diet).



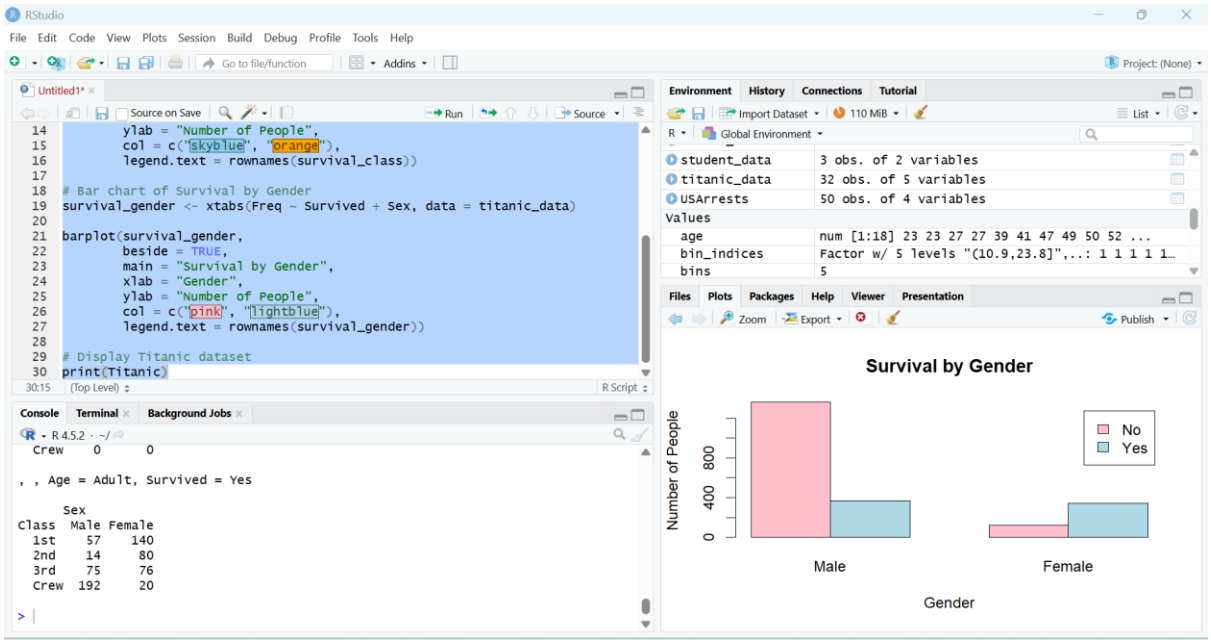
15. Perform EDA on iris dataset: dimensions, summary, standard deviation,quantiles, grouping by Species, pivot table, categorical grouping with Sepal.Length categories.



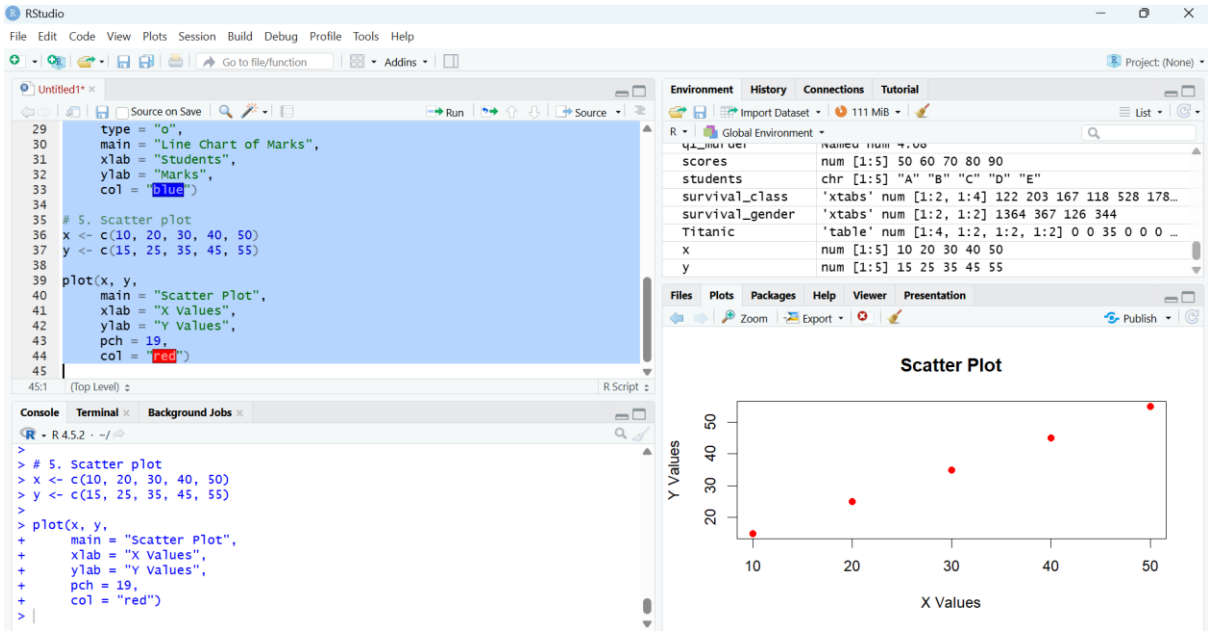
16. Explore USArrests dataset: summary statistics, state with largest rape arrests, max & min murder rates, correlation among features, states above median assault arrests and bottom 25% for murder, visualization with histogram, density, scatterplots, bar graphs.



17. Explore Titanic dataset: bar chart of survival vs class, modify plot by gender, histogram of Age.

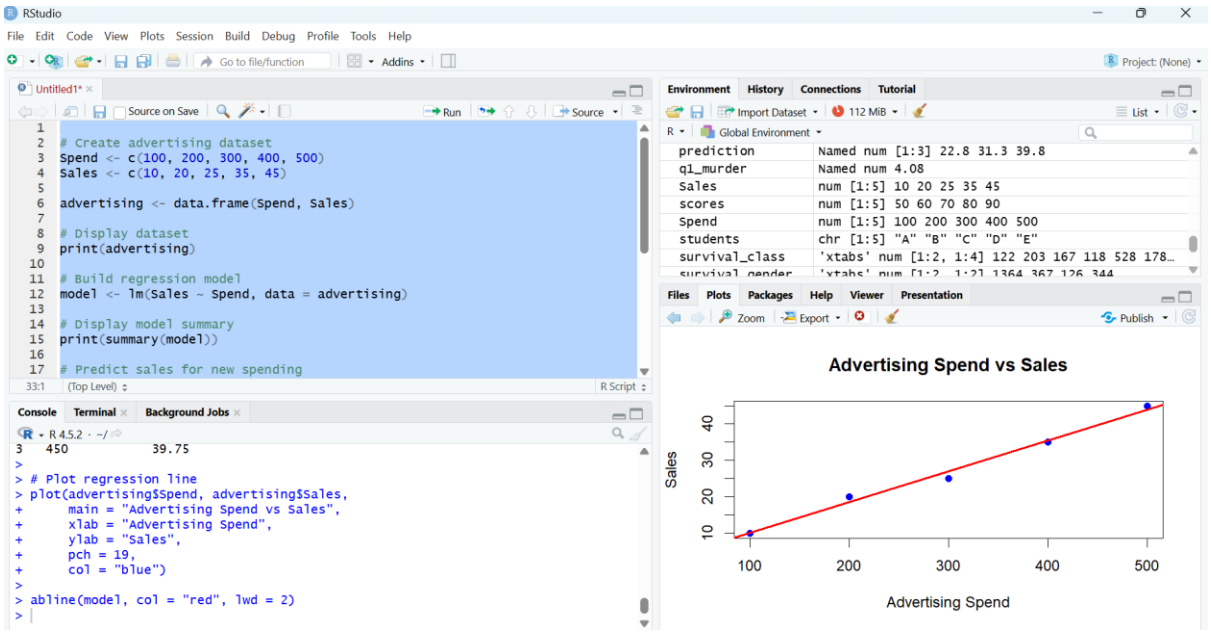


18. Create graphs in R: boxplot, histogram, bar plot, line chart, scatter plot.

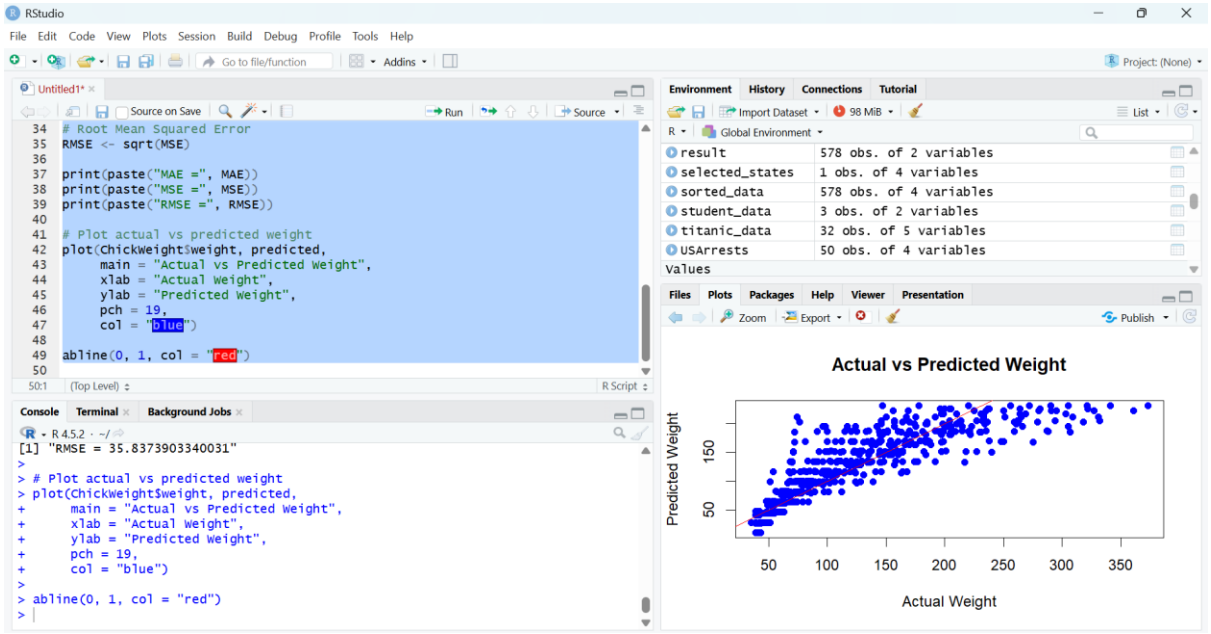


19. Build a regression model on advertising dataset (Sales ~ Spend) and predict

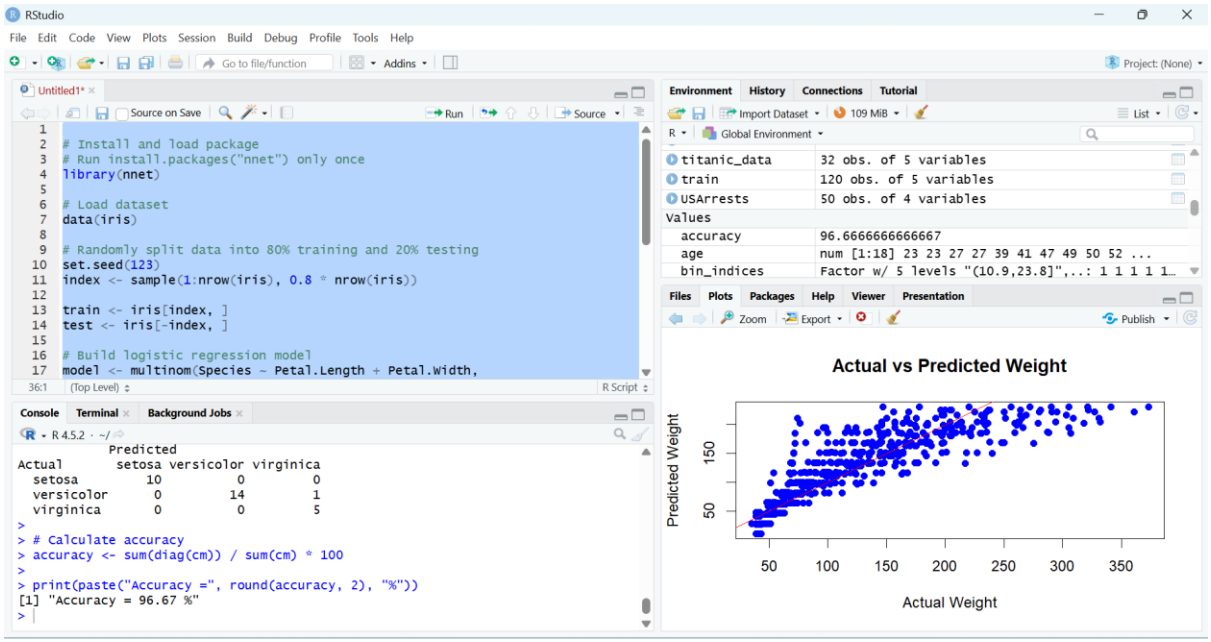
Sales.



20. Create multiple regression model using ChickWeight dataset with “Time” and “Diet” as predictors; predict weight and compute model error.



21. Randomly split iris dataset into train/test (80/20), build logistic regression (Species ~ Petal.Length + Petal.Width), predict, and evaluate with confusion matrix.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
Source on Save
Run
Source
1 # Install and load package
2 # Run install.packages("nnet") only once
3 library(nnet)
4
5 # Load dataset
6 data(iris)
7
8 # Randomly split data into 80% training and 20% testing
9 set.seed(123)
10 index <- sample(1:nrow(iris), 0.8 * nrow(iris))
11
12 train <- iris[index,]
13 test <- iris[-index,]
14
15 # Build logistic regression model
16 model <- multinom(Species ~ Petal.Length + Petal.Width,
36:1 (Top Level) R Script
Console Terminal Background Jobs
R 4.5.2 · ~/
Predicted
Actual setosa versicolor virginica
setosa 10 0 0
versicolor 0 14 1
virginica 0 0 5
> # Calculate accuracy
> accuracy <- sum(diag(cm)) / sum(cm) * 100
> print(paste("Accuracy =", round(accuracy, 2), "%"))
[1] "Accuracy = 96.67 %"
Environment History Connections Tutorial
Import Dataset 109 MiB
List
R Global Environment
titanic_data 32 obs. of 5 variables
train 120 obs. of 5 variables
USArrests 50 obs. of 4 variables
Values
accuracy 96.6666666666667
age num [1:18] 23 23 27 27 39 41 47 49 50 52 ...
bin_indices Factor w/ 5 levels "(10.9,23.8]",..: 1 1 1 1 1...
Files Plots Packages Help Viewer Presentation
Zoom Export Publish
Actual vs Predicted Weight
Predicted Weight
50
150
50 100 150 200 250 300 350
Actual Weight